AWANTIKA SRIVASTAVA

AI/ML Engineer | LLMs & AWS

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PROFILE SUMMARY

AI/ML Engineer with 2+ years of experience building and deploying scalable machine learning and generative AI solutions. Skilled in Python, TensorFlow/PyTorch, and cloud deployment on AWS with hands-on experience in RAG pipelines, deep learning, and production ML systems. Proven ability to design, optimize, and integrate AI models into real-world applications.

CORE TECHNICAL SKILLS

- **Programming & Backend:** Python, C++, SQL., FastAPI, Flask, REST APIs, Microservices Architecture
- **Statistics & Mathematics:** EDA, Statistical Modeling, Descriptive & Inferential Statistics, Bayesian Statistics, Hypothesis Testing, Probability, A/B Testing
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, LightGBM, Model Evaluation Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC).
- **Deep Learning:** ANN, CNN, RNN, LSTM, BERT, Transfer Learning, Self-Attention, multi-head Attention, Tensorflow, PyTorch, Keras
- **NLP:** Text preprocessing, Tokenization, Chunking, Sentiment Analysis, Transformer-based models, NLTK, SpaCy, Time-Series Forecasting, ARIMA, SARIMA
- **Generative AI & LLM Systems:** RAG, Multi-Agent Systems, Agentic AI, Prompt Engineering, LLM Fine-Tuning (LoRA), Vector DB, Embeddings
- **Computer Vision:** Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Video Analytics, OpenCV, Paddle OCR
- **MLOps & Deployment:** Docker, MLflow, CI/CD Pipelines, Model Monitoring, Production Deployment
- **Cloud Platforms:** AWS (EC2, S3, DynamoDB), Cloud-based ML deployment
- **Tools & Frameworks:** LangChain, LangGraph, HuggingFace, FAISS, ChromaDB, Git, Streamlit

EXPERIENCE

AI/ML Engineer | PPS International Pvt. Ltd. **January 2024-Present**

- Designed and deployed **scalable AI/ML models for predictive analytics and automation systems** using Python and machine learning frameworks.
- Built **production-ready ML pipelines** for model training, evaluation, and deployment using Spark and PySpark.
- Developed **classification and regression models** using Random Forest, Gradient Boosting, and deep learning architectures, improving prediction accuracy by **15–20%**.
- Designed **LLM-based intelligent systems using Retrieval-Augmented Generation (RAG)** for knowledge retrieval and automated query handling.
- Implemented **multi-agent workflows and tool-calling pipelines** for automated reasoning and task execution.
- Fine-tuned large language models using **LoRA / PEFT techniques** to improve domain-specific response quality.
- Integrated **vector search systems (FAISS, Elasticsearch)** for semantic search and document retrieval.
- Deployed AI models on **AWS infrastructure using Docker-based containerized environments** ensuring scalability and low-latency inference.
- Managed **model lifecycle and experiment tracking using MLflow**.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- **Designed and deployed a** real-time computer vision–based ML system **for unsafe** driver behavior detection using **CNN-based SSD MobileNet** model.
- Trained and optimized models on large-scale video datasets, achieving **20–25 FPS** real-time processing with **<150 ms inference latency**.
- Implemented **end-to-end ML pipelines** for data ingestion, preprocessing, model training, evaluation, and production inference.
- Deployed optimized models using **TensorFlow Lite** on edge/production environments for continuous monitoring.
- Built a **Flask-based web dashboard** to visualize detections and automatically record **30-second event clips**, reducing manual review effort.

Chatbot Using LLM & RAG | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Applied **prompt engineering techniques** to improve response relevance and consistency.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

YouTube Comments Sentiment Analyzer | Link-https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Developed **LSTM-based time-series forecasting** model trained on 5+ years historical stock data reducing **RMSE** by 18**%.**
- Engineered lag **features**, **rolling statistics**, and volatility indicators improving prediction stability.
- **Implemented** custom **BPTT training** loop with adaptive learning rate scheduling for **stable convergence**.
- Applied **regularization techniques** reducing overfitting gap by **20**%.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology